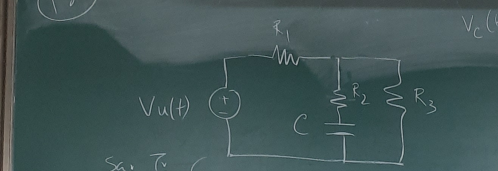
PS1



$V_c(t)$

$V_c(\infty) = \frac{V}{R_1+R_3} R_3$

t

$t \to \infty$

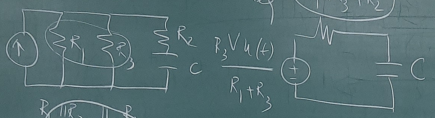
$V' = \frac{V R_3}{R_1+R_3}$

$V'\left(1 - e^{-\frac{t}{R_{eq}C}}\right)$

Sia τ_2

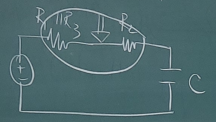
$u_c(0_-) = 0$

$R_{eq} = R_1 \| R_3 + R_2$

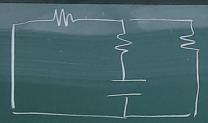


Driven

$R_1 \| R_3 \quad \frac{V u(t)}{R_1}$



PS2



$V_c(0_-) = V_0$

$V_c(t)$

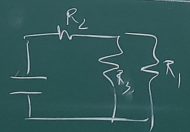
$V_0 \, e^{-t/R_{eq}C}$

$RC\frac{dv}{dt} + V = k$

First order systems

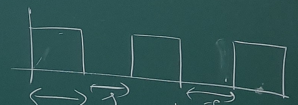
Natural response

Transient



L ? Same trick!

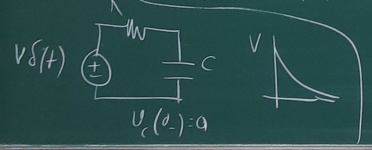
$\frac{L}{R_{eq}}$



> time-constant

pulse

$\delta(t)$



p53

$i(t)$, $t \geq 0$

L, C

$v_c(0) = V$

Second order system

SL, $\frac{1}{sC}$, $\frac{V}{s}$

$$I(s) = \frac{\frac{V}{s}}{sL + \frac{1}{sC}} = \frac{1}{L}\frac{V}{s^2 + \frac{1}{LC}} = \frac{1}{L}\frac{V}{s^2 + \left(\frac{j}{\sqrt{LC}}\right)^2} = \frac{1}{L}\frac{V}{\left(s + \frac{j}{\sqrt{LC}}\right)\left(s - \frac{j}{\sqrt{LC}}\right)} = \frac{V}{L}\frac{1}{\frac{2j}{\sqrt{LC}}}\left(\frac{1}{s - \frac{j}{\sqrt{LC}}} - \frac{1}{s + \frac{j}{\sqrt{LC}}}\right)$$

$$i(t) = \mathcal{L}^{-1}\left(I(s)\right)$$

$x^2 - a^2$

$$I(s) = \frac{V}{j}\sqrt{\frac{C}{L}}\frac{1}{2}\left[e^{\frac{jt}{\sqrt{LC}}} - e^{-\frac{jt}{\sqrt{LC}}}\right] u(t)$$

(Laplace)

$$\frac{1}{s+a} \Rightarrow e^{-at}u(t)$$

= Sinusoidal

$2\pi f = \omega = \frac{1}{\sqrt{LC}}$

p54

$$L\left(\frac{di}{dt}\right) + V_c = 0$$

$$i = C\cdot\frac{dV_c}{dt} \Rightarrow L\frac{d^2V_c}{dt^2} + V_c = 0$$

$Q = CV_c$

$i = \frac{dQ}{dt} = C\frac{dv_c}{dt}$

2nd order

⇒ Ideal circuit

In practice,

R, L, C, $v_c(0) = V$

{ res of inductor, coil, wires, etc.

$$I(s) = \frac{V/s}{sL + \frac{1}{sC} + R} = \frac{VC}{s^2LC + sRC + 1}$$

r_1, r_2

$$= V_C\left[\frac{A}{s - r_1} + \frac{B}{s - r_2}\right]$$

PSS

$$r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sridharacharya's formula

$$ax^2 + bx + c = 0$$

$$r_1, r_2 = \frac{-RC \pm \sqrt{R^2C^2 - 4LC}}{2LC}$$

Case I will happen if

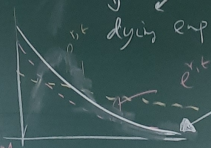
$$R^2C^2 > 4LC \Rightarrow \sqrt{R^2C^2 - 4LC}$$

Dominating eq: $\boxed{-RC + \sqrt{RC - 4LC^2}} < RC$

Case I: Overdamped (r_1, r_2 -ve & real)

$i(t) \propto A e^{r_1 t} + B e^{r_2 t}$, $r_1, r_2 < 0$ real

$r_1 t$ & $r_2 t$ dying exp



No hunt of sinusoid

$$\lambda_2 < \lambda_1 \qquad e^{-\lambda_1 t} + e^{-\lambda_2 t} = e^{-\lambda_2 t}\left(1 + e^{-(\lambda_1 - \lambda_2)t}\right), \lambda_1, \lambda_2 > 0$$

$= 1$

Case II: Underdamped (some sinusoid)

$i(t) \propto A e^{r_1 t} + B e^{r_2 t}$ (ignore V_C)

We get sinusoid only if r_1 or r_2 has imaginary parts!

↳ ensured if $R^2C^2 < 4LC$

$$\left(-\frac{RC}{2LC}\right) \pm \sqrt{\frac{R^2C^2 - 4LC}{(2LC)^2}}$$

$$A e^{-\sigma t} e^{j\omega_d t} + B e^{-\sigma t} e^{-j\omega_d t}$$

σ = damping Fac

$j\omega_d$ = Sinusoid w/ decaying amp

$$= e^{-\sigma t}\left(A e^{j\omega_d t} + B e^{-j\omega_d t}\right)$$